

# PhaseShift Technologies — Venture Brief (provisional, 5 pages)

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*Provisional 5-page format. Figures follow our business plan v1.9 (Aug 29, 2026) and the price revision of Sep 1, 2026. "Estimate" = our calculation from public data; "not yet measured" = open. Customer names are withheld (Producer A, Producer B) because those discussions are confidential.*

## 1. Summary

**PhaseShift recovers iodine from the dilute wastewater that today's iodine plants discharge, using a temperature-responsive polymer that captures iodine at single-digit ppm and releases it on a temperature swing.** The unit is a side-stream retrofit that the producer owns; we earn recurring revenue from polymer supply and a per-kilogram process fee on the iodine recovered.

|              |                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Origin       | Doshisha University (Kyoto), Faculty of Science and Engineering. PI: Assoc. Prof. Shinnosuke Nishimura (inventor).                                        |
| Stage        | Laboratory proof-of-concept, batch scale (TRL3). TRL4 assessment on real wastewater from two domestic producers scheduled for late September 2026.        |
| Beachhead    | Iodine producers on the Boso gas field, Chiba, Japan (about 80% of Japan's output; Japan supplies about 30% of the world's iodine).                       |
| Global path  | Produced-water operators in Oklahoma's Anadarko Basin (Phase 2, gated on domestic results).                                                               |
| IP           | Foundational polymer patent application published (Doshisha University). First application patent scheduled for filing October 2026.                      |
| Team         | Business lead: Junichi Kato (five foundings, cumulative JPY 450M seed raised). Technical Advisor / PI: Assoc. Prof. Nishimura. One research staff member. |
| Funding plan | Two-stage J-KISS (JPY 70M angel, JPY 120M seed). JST D-Global 3-year R&D application due Sep 30, 2026.                                                    |
| Ask of GSC   | US customer discovery in produced water, business-model stress test, US/PCT IP strategy, investor readiness, third-party validation partners.             |

## 2. The problem: iodine that is produced but never collected

Japan is the world's second-largest iodine producer. The iodine comes from natural-gas brine pumped in Chiba Prefecture and is recovered by two commercial methods, blow-out and ion exchange, which work at raw-brine concentrations (tens of ppm to about 100 ppm) and become uneconomic below that.

From published operating data for a representative plant (35 ppm brine, 60,000 t/day, 1,400 kg/day of iodine), we estimate that only about two-thirds of the iodine entering the plant is recovered. The rest, roughly 700 kg/day at that plant and, by our estimate, **700–1,000 t/yr nationwide**, leaves with the discharged brine at 7–12 ppm. One major producer has told us its discharge carries about 10 ppm, consistent with our estimate. (The national figure is being cross-checked with the Society of Iodine Science; until then it is our estimate.)

Producers cannot pump more brine: pumping is capped by ground-subsidence agreements with local governments. **Recovering iodine from their own discharge is therefore one of the few ways left for a Japanese producer to increase output.** That is what we sell: more iodine from the same brine, with the existing process untouched.

**Where iodine goes, and why demand is not the constraint.** By volume, iodine is used in X-ray contrast media, pharmaceuticals, LCD polarizing film, iodophor disinfectants, animal feed and fluorochemicals, in that order (USGS); spot prices averaged around USD 70/kg in 2024–2025. Two newer uses add a supply-security dimension. Japanese industrial policy names iodine a principal raw material for perovskite solar cells, which Japan intends to manufacture domestically at scale. In space, iodine is displacing xenon as an electric-propulsion propellant: the first in-orbit demonstration of an iodine thruster was published in *Nature* in 2021, and iodine stores as an unpressurised solid at roughly three times xenon's density, which simplifies small-satellite systems. Xenon output is concentrated (Russia and Ukraine, about a quarter to a third of world supply) while third-party projections put more than 20,000 satellites in orbit within a decade, most requiring electric propulsion. **Propulsion volumes are small today against a roughly 34,000 t/yr iodine market, and we put none of this in our financials.** The point is directional: two industries that matter to both governments now draw on the same constrained primary supply, and Japan is the second-largest source of it.

### 3. Technology and evidence

#### 3.1 How it works

The core material is a temperature-responsive polymer (the "PhaseShift agent") from the Nishimura laboratory, Doshisha University. In water it complexes with iodine and, on a modest temperature change, phase-separates into a concentrated iodine-rich phase; a further temperature swing releases the iodine. Desorption is therefore **designed** to need no reducing agents or solvents, and the heat source is **designed** to be low-grade plant waste heat (one producer has disclosed a steam-drain stream at 80–90 °C). Only the concentrated phase, about 1–3% of the flow, is heated; we estimate about 10.5 kW per unit. Whether waste heat alone suffices depends on heat-exchanger efficiency and is not yet demonstrated.

Installation is a **side-stream retrofit**: 3.5–6.8% of the discharge is diverted through a fixed-bed cartridge unit and returned; the main line never stops. This followed producer feedback that full-flow insertion was not feasible.

#### 3.2 What has been measured (laboratory, batch, as of August 2026)

| System                             | Conventional polymer (PVP) | PhaseShift agent                  | Note                                         |
|------------------------------------|----------------------------|-----------------------------------|----------------------------------------------|
| Iodine in hexane                   | 73.9%                      | 91.4%                             | capture                                      |
| Pure water, 7 ppm (two conditions) | —                          | 96.8% / 97.2%                     | capture, UV-vis 351 nm, supernatant basis    |
| Model brine wastewater, 7 ppm      | —                          | supernatant below detection limit | capture                                      |
| Reuse, second cycle                | —                          | 79.4%                             | dose not recorded; cause under investigation |

These are **capture** figures derived from the drop in supernatant absorbance. They do not include desorption and collection. **Recovery on real wastewater has not yet been measured.** Our business plan uses a recovery rate placeholder of 92.9%, which we flag wherever it drives a number.

#### 3.3 Known gaps (disclosed at application)

1. **Polymer loss of about 30% per cycle** (up to 50%) in the current liquid-liquid separation, because the dilute phase retains dissolved polymer; this bears directly on consumables economics. A copolymer that precipitates as a solid (gravity settling instead of centrifugation) has been made; its loss rate is not yet measured.
2. **No continuous-flow data.** The lab protocol uses static settling and centrifugation; a producer has named continuous flow its heaviest requirement. A continuous-flow unit is the first engineering task.
3. **Iodine speciation.** Discharge may carry residual reducing agent, leaving iodide rather than iodine; pre-oxidation requirements are being defined.
4. **High salinity.** Phase-separation temperature shifts with salt (38 °C in water, 35 °C in brine). Behavior above 100,000 mg/L TDS, typical of US produced water, is untested.

### 4. Market and customers

#### 4.1 Phase 1 — Japan (beachhead)

| Item                                                   | Value                                                                 | Basis                                                                                                                 |
|--------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Uncaptured iodine, national                            | 700–1,000 t/yr                                                        | estimate from published plant data; being cross-checked                                                               |
| Value at our net price (JPY 8,587/kg)                  | JPY 6.0–8.6 bn/yr                                                     | net-to-gross ratio under review                                                                                       |
| Addressable by our model (Chiba, top-5 municipalities) | JPY 2.95–4.21 bn/yr                                                   | Chiba 82% of national output × 59.8% in top-5 municipalities                                                          |
| Per site (design case)                                 | 4,316 kg/yr recovered; JPY 50.9M/yr additional sales for the customer | 2,074 m³/day side stream, 320 days, 7 ppm, recovery placeholder 92.9%, gross price JPY 11,782/kg (USD 74.27 × 158.64) |
| Sites by FY2030                                        | Base 5 (downside 3, upside 7–9)                                       | business plan v1.9                                                                                                    |

Customers are the producers themselves. Two of Japan's major producers are in technical discussions with us: Producer A under a three-party NDA since August 2026, with real wastewater samples agreed; Producer B since a first technical

meeting on Aug 27, 2026, with samples agreed and a written list of adoption requirements (selective adsorption in brine, easy desorption, continuous-flow configuration). Chiba Prefecture has arranged a first meeting with a third producer for mid-September. One producer reports discharge of up to 18,000–19,000 m<sup>3</sup>/day, about nine times one unit's 2,074 m<sup>3</sup>/day design flow.

## 4.2 Phase 2 — North America (conditional)

US iodine comes largely from oil-and-gas produced water in Oklahoma's Anadarko Basin, where Iofina operates eight plants (each 10,000–50,000 bbl/day) and has announced a Permian plant for late 2026. Back-calculating from disclosed output, we estimate these plants take water at **59–108 ppm**. Two customer groups follow:

- **Downstream of existing plants.** If plant recovery is 70–90%, residual streams run 6–33 ppm, comparable to or richer than Japan's tails.
- **Streams that never reach a plant.** Water-midstream and saltwater-disposal (SWD) operators aggregate produced water (one operator alone reports 60 disposal facilities and 1.2 million bbl/day of capacity). Below the concentration at which iodine value equals disposal cost (**51–106 ppm** at USD 74.27/kg and disposal costs of USD 0.60–1.25/bbl), no one builds a dedicated plant. A low-CAPEX retrofit owned by the water operator is the only form that works there.

One Japanese design site (about 13,000 bbl/day) sits at the lower end of a US plant intake, so the unit transfers without redesign. Two open questions decide Phase 2 and are what we want to test through GSC: **do low-concentration streams exist at meaningful volume at aggregation points**, and **does the polymer work above 100,000 mg/L TDS**. If either answer is no, Phase 2 stops. Iodine is not on the US critical minerals list; we do not build the case on subsidies.

## 5. Business model and unit economics

Three revenue layers, all keyed to a unit the customer owns and operates:

| Layer                              | Price (Sep 2026)                    | Nature                   |
|------------------------------------|-------------------------------------|--------------------------|
| 1. Unit sale                       | JPY 100M per unit (about USD 0.63M) | one-time; customer CAPEX |
| 2. Consumables and maintenance     | JPY 10M per year per unit           | recurring polymer supply |
| 3. Process fee on recovered iodine | JPY 2,000 per kg-I <sub>2</sub>     | recurring, volume-linked |

**Customer P&L per site, per year (design case):** additional iodine sales +JPY 50.9M; consumables –JPY 10.0M; process fee –JPY 8.6M; on-site power –JPY 0.6M; **net additional gross profit +JPY 31.6M**; simple payback **3.2 years** (about 2.2 years with Japanese investment tax incentives). All of these scale with the unverified recovery rate.

**Why asset-light is a consequence, not a preference.** Below the break-even concentration, iodine returns only part of the disposal cost, so a vertically integrated plant cannot pay for itself; the only design that works is a low-cost retrofit on existing assets, paid for by the asset owner and monetized through consumables and a volume fee.

**An unfavorable number we disclose.** Per kilogram of annual capacity our unit costs about USD 146 against about USD 44 for Iofina's newest plant, roughly 3.3×, because we process much thinner water. The gap narrows if North American tails are 20–30 ppm (3–4× output per unit) and is offset, in the customer's view, by the payback above.

**Company revenue (illustrative, FY2030, 5 sites):** about JPY 293M across the three layers. Cost of goods is being rebuilt after the September price revision, so we do not quote operating profit.

## 6. Competition and intellectual property

| Alternative                                                     | Position                                                              | Our reading                                                                                           |
|-----------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Blow-out and ion exchange (incumbent)                           | Economical at higher concentrations; used by all Japanese producers   | Neither side has published data at 7–12 ppm. We do not claim the band is uncontested.                 |
| Iofina IOsorb (USA)                                             | Vertically integrated, 8 plants, 59–108 ppm feed (estimate)           | We sit downstream, not in competition for the same water. Risk: Iofina moves to lower concentrations. |
| WO2025258648A1 (a Japanese university with a domestic producer) | Adsorbent for ≥10 ppm; desorption with reducing agent                 | Closest patent. Ours desorbs by temperature swing; regeneration data for both are limited.            |
| Reverse osmosis                                                 | Rejection collapses at low feed (about 10% below 15 ppm in one study) | Not viable in our band.                                                                               |

|            |                      |                                                                      |
|------------|----------------------|----------------------------------------------------------------------|
| Do nothing | Discharge or dispose | The real alternative. We add revenue rather than replacing disposal. |
|------------|----------------------|----------------------------------------------------------------------|

**IP.** A foundational patent application covering the polymer family is filed and published in Japan (Doshisha University, sole applicant; inventor: the PI). Our patent attorney has confirmed that a design outside the foundational claims is feasible; its structure is withheld until the **first application patent** (joint filing, Doshisha University and Kato; invention committee approval Aug 7, 2026) is filed in **October 2026**. Information is managed in three layers: public (function, targets, customer value), demonstration-required (operating conditions, analytics, connection specs) and core (formulation, manufacturing, control logic, cartridge internals). University-led public R&D results stay with the university under an exclusive option; company-led R&D is contracted so new IP is solely owned.

## 7. Global strategy and gates

| Phase                | Scope                                                             | Gate to proceed                                                                                                                                                            |
|----------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (now–2028)         | Japan: TRL4 → paid field unit at a producer                       | TRL4 result (Sep 2026); paid demonstration or equivalent LOI                                                                                                               |
| 2 (earliest 2027 Q2) | 12-week North America feasibility study, then field demonstration | Nine conditions incl. domestic TRL4 pass, application patent filed, ring-fenced funding, a US prime partner candidate, a site candidate and a third-party verification lab |

The order is fixed: North America is where a proven domestic design goes next, not a parallel bet. The GSC sprint sits between the two phases: it lets us answer the two Phase-2 questions and fill the partner-related gate conditions before committing money, without slowing domestic work.

## 8. Team and partners

**Junichi Kato — Business lead / CEO candidate.** Serial entrepreneur, five foundings. Built a Web3 solutions company in Malaysia to an 80-person multinational team; CEO of a Los Angeles-based satellite-data venture; cumulative JPY 450M seed raised. Since 2023, regional business development with Chiba local governments; Regional Revitalization Entrepreneur, Shirako Town. Full-time on PhaseShift since April 2026. University of Tokyo Matsuo-Iwasawa Lab AI Management Course; NEDO SSA.

**Shinnosuke Nishimura, Ph.D. — Principal researcher; Technical Advisor to the venture.** Associate Professor, Doshisha University. Inventor of the foundational patent. Has agreed to serve as PI for the JST D-Global application.

**Research staff (1).** Materials synthesis and process validation.

**Support network.** Doshisha startup-support coordinator (IP confirmation, contracts); patent attorney and INPIT specialist (neutral review); CIC / Chiba Prefecture startup program (monthly); Chiba Prefecture industry division (producer introductions); TIB STUDIO (Tokyo). Recruiting: CxO candidates for technology and finance, and a polymer manufacturing / unit engineering partner.

## 9. Milestones and funding

| When              | Milestone                                                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2026          | TRL4 assessment on real wastewater (Producer A, Producer B; samples agreed) with a predefined go / conditional-go / pivot / stop rule. JST D-Global application (3-year R&D, FY2027–29) with a commercialization-promotion organization. |
| Oct 2026          | First application patent filed.                                                                                                                                                                                                          |
| Oct 2026–Feb 2027 | GSC sprint; Silicon Valley bootcamp Nov 30–Dec 11.                                                                                                                                                                                       |
| 2027              | 100 L/day bench unit (TRL5): mass balance, consumable life, waste-heat energy balance.                                                                                                                                                   |
| 2028              | On-site pilot at a producer, side-stream (TRL6).                                                                                                                                                                                         |
| 2029              | North American sample validation and partner framework (TRL7); go / no-go on Phase 2 field demonstration.                                                                                                                                |

**Funding.** Two-stage J-KISS: JPY 70M angel, then JPY 120M seed (JPY 190M total). Incorporation is tied to public-funding outcomes rather than a fixed date: if D-Global is awarded, R&D runs at the university first and the company is formed at the end of that period; if not, immediately. An ERCA grant (JPY 4M) funds part of the 2026 work. Unit economics assume no subsidies.

## 10. What we want from GSC, and how we will know the sprint worked

1. **US customer discovery in produced water** — introductions to water-midstream / SWD operators and produced-water experts in Oklahoma and Texas. *Success*: a concentration-and-volume map of candidate aggregation points and two operators willing to share water-quality data.
2. **Business-model stress test** — mentors who have commercialized separation or water technologies, including offtake for concentrated iodine (US water operators, unlike Japanese producers, do not refine). *Success*: a US-specific model with named offtake options.
3. **IP and regulatory strategy** — US/PCT filing plan for the October 2026 application, freedom to operate versus Iosorb-type resins, produced-water rules. *Success*: a filing decision and an FTO memo.
4. **Investor readiness** — feedback from deep-tech and climate investors on the milestones that make a Japan-first resource-recovery venture fundable. *Success*: a seed narrative and target list for 2027.
5. **Third-party validation partners** — a US university or lab for high-salinity tests. *Success*: one signed test plan.

We bring real wastewater data from Japanese producers, a numbered gate framework, and a business lead who has built and raised in Southeast Asia and the US and works directly with mentors and investors in English.

## 11. Risks and open questions

| Risk                                                                             | Our response                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRL4 cannot be run in September or lands in the "hold" band                      | Quantification route for low ppm being secured (ion chromatography, external labs); decision timing slips to October rather than the plan being changed.                                                                   |
| Recovery rate (placeholder 92.9%) comes in low                                   | All site-level figures are recomputed; exit thresholds are pre-set and will not be loosened without new evidence.                                                                                                          |
| Polymer loss stays near 30%; waste heat insufficient                             | Solid-precipitating copolymer, else consumables pricing and contract structure are redesigned. Reduce heated fraction, direct heat exchange at the separator, or auxiliary heat; if all fail, re-price on purchased power. |
| Incumbents raise their own recovery and tails shrink                             | Establish operating data and installed base first; bromine co-recovery from the same brine as a later option (not in the financials).                                                                                      |
| Polymer fails above 100,000 mg/L TDS                                             | Phase 2 stops or shifts to lower-TDS streams.                                                                                                                                                                              |
| Iofina moves down in concentration; separation mechanism alone is a weak barrier | Ownership model and installed data in Japan; downstream positioning in the US; barrier moved to operating data, installation know-how and consumable supply.                                                               |

*End of brief.*